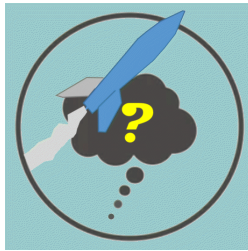

Richard Nakka's *Experimental Rocketry* Web Site

Introduction to EX Rocket Design



4. Rocket Body Design Considerations

Aerodynamic Loading

A good starting point in designing a rocket is to decide how big we want our rocket to be, specifically, the body diameter. And how this relates to our goals. The diameter of the rocket body has a particularly strong influence on how high a rocket can fly with a given motor. This is because drag force (F_d) acting against a rocket's upward motion is proportional to the *body diameter squared*. The equation for rocket aerodynamic drag was given earlier by Equation 1 (or [Equation 2](#), for supersonic flight, however, read this [note](#)):

$$F_d = \frac{1}{2} \rho V^2 C_d A \quad [\text{Equation 1}]$$

ρ (Greek: rho) = air density

V = velocity

C_d = drag coefficient

A = frontal area of rocket, where $A = \frac{1}{4} \pi D^2$

Doubling the diameter (D) of the rocket body increases drag force by a factor of *four*. Or put another way, reducing the body diameter by $\frac{1}{2}$ reduces drag losses to $\frac{1}{4}$. Aerodynamic drag losses can be very significant, especially at high velocity, and can reduce peak altitude by 50% (compared to zero drag condition) or more. So to achieve an altitude goal with a minimum sized motor, the body tube should be made as *small a diameter as practical*. In other words, if your motor, recovery system and payload can readily fit within a 2-inch (50mm) body, use that rather than a 3-inch (75mm) body. Volume for payload and the recovery system can be gained by adding *length* to the rocket. A side benefit to a longer rocket is greater stability, due to centre of gravity tending to be being further forward. A note of caution. As was demonstrated earlier, *body frictional drag* typically accounts for the largest single drag component of a rocket, so a longer rocket should be made with attention to providing a smooth finish.

If designing a rocket for a particular motor, the diameter of the rocket motor will influence the size of the body tube. Nearly all the rockets that I have constructed have had the motor fit within the body tube. It is possible to have the motor serve as airframe, but this is not normally done due to the complexities that result, such as suitably mounting the fins (this is doable, here is an [example](#)). For exceptionally large rockets that are intended to maximize altitude performance, this is a good option, however.

A rocket body, in particular the choice of diameter, wall thickness and material, can comprise a significant percentage of a rocket's total mass, typically 30% or more. Care should be taken in the selection of all three of these parameters. Doubling a rocket body's diameter will double its mass (on top of the drag penalty already discussed). Same is true for wall thickness. Choice of material has a far less clear influence on mass. Aluminum has twice the mass density of fibreglass or a polymer material such as PVC. However, aluminum has much greater strength and it is possible to have an aluminum body with a thin wall thickness. For example, my *Xi* rocket has an aluminum body of 3 inch (76mm) diameter and a wall thickness of 0.035 inch

(0.9mm) and as such is lightweight. The drawback, however, is cost. This tubing has a cost of \$18 USD per foot.

Another factor to consider with regard to body tube selection is strength. In flight, a rocket body is subjected to compression and bending loads. Compression load is a result of drag force acting on the nosecone and to a lesser extent frictional drag on the body surface. Inertia loading due to mass of the body tube (and anything attached to it such as nosecone and instrumentation) acting under acceleration must also be considered. Bending load is a consequence of non-zero angle of attack of the flight path. This may be a result of wind gusts acting perpendicular to the rocket, slightly misaligned fins, or protruberances such as launch lugs or camera fairings. Another potential cause of non-zero angle of attack is dynamic instability (or imbalance). If a rocket rolls (spins), by even a small amount, **coning motion** of the rocket will almost certainly result. This is a result of a rocket's principal axis of inertia not being exactly in line with its geometric axis, a consequence of components housed within the rocket (such as payload) being located offset from the rocket's longitudinal axis. This important topic is discussed in detail in the [Dynamic Stability](#) webpage.

For a high performance rocket that achieves supersonic velocity, due diligence is wise, as drag force increases dramatically as a rocket approaches and exceeds mach one. Figure 1 illustrates an example of how the drag coefficient of an example rocket can dramatically increase as Mach one is approached and exceeded.

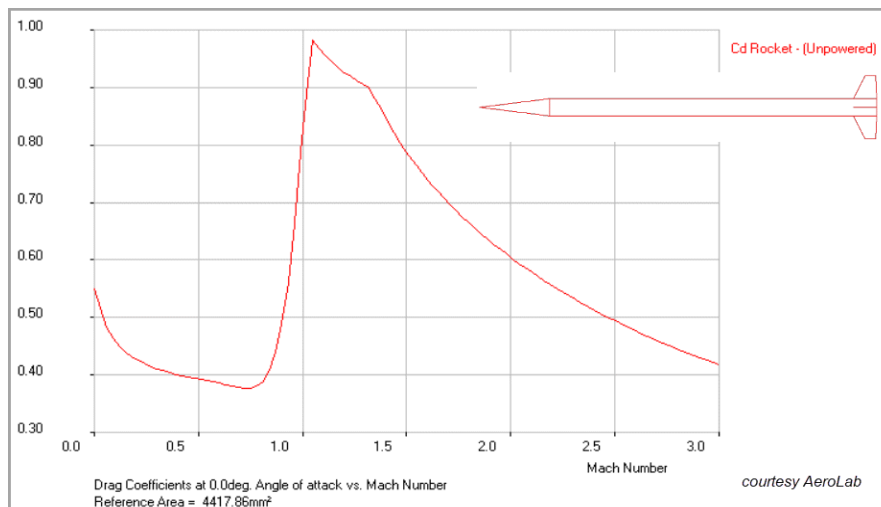


Figure 1: Drag coefficient versus mach number for an example supersonic rocket

As is apparent from equation [1] and [2], drag force is directly proportional to drag coefficient. As the supersonic rocket in this example goes from a velocity of mach 0.5 ($C_d=0.4$) through to mach 1.1 ($C_d=0.98$), the drag coefficient (and therefore drag force) increases by a factor of nearly 2.5. As such, for an EX rocket that is expected to "break the sound barrier" an assessment of the strength of the body due to aerodynamic loading is recommended.

Bending Loads

Being long, slender and of lightweight construction, a typical EX rocket is most vulnerable to stresses resulting from *bending*. This is especially true if a rocket is made of sections, connected by joints, as most EX rockets are. Bending loads on a rocket body can be expected due to:

1. Handling/abuse on the ground or during transport
2. Aerodynamic flight loads
3. Flight loads other than aerodynamic, such as chute deployment event or touchdown

Ground Loads

Bending loads acting on the rocket body can occur during "handling" or during transport to the launch site. For example, you might be carrying the rocket to the launch pad, holding it in the middle (or at the balance point) in a horizontal orientation. A larger rocket might be transported horizontally, supported in a *cradle*. Such a cradle, if properly designed, will effectively support a rocket such that undue bending loads are not expected during transport. What may not be expected could result when the rocket is removed (or installed) in the cradle. The rocket may be simultaneously lifted at the forward end (nosecone) and at the aft end. The dead

weight of the rocket will induce an unexpected bending load. This is known as "abuse loading" which is something that may be taken into account in the design. Fortunately, round tubing is an inherently strong structural shape and most commercial off-the-shelf tubing that is readily available, and suitable for a rocket body, is sufficiently robust that structural failure during handling is unlikely. Joints, however, may not be as strong as the basic tubing and could serve as a weak point if not properly designed.

As such, it is prudent to design a rocket to ensure handling loads will not damage the rocket assembly. It may also be prudent to *proof test* the rocket body. A *proof test*, as the name implies, demonstrates proof that a rocket, as-designed and as-built, meets the strength requirement that will help assure it will endure the rigours of flight loads and handling loads. This will be discussed later.

Flight Loads

If a rocket is launched in windless conditions and if the rocket is aerodynamically balanced (such that it flies at zero angle of attack), and the thrust vector is in line with the rocket's long axis, the rocket will not experience any bending loads during its ascent. The most significant bending loads acting on a rocket in flight are generated as a result of sudden wind gusts or a result of dynamic imbalance. Both result in non-zero angle of attack of varying severity. Gusts, acting horizontally, generate *normal* forces on the rocket body. The term *normal* refers to loads that are applied perpendicular to the rocket's long axis, or side loads. A sudden gust velocity vector (U), when added to the vehicle's velocity vector (V) causes a small [angle of attack](#) (angle of attack is symbolized by the Greek letter alpha, α) to the rocket's flight path.

Flying at some non-zero angle of attack, whether caused by a gust or by dynamic imbalance of the rocket, generates normal forces acting on the nosecone, fins, transitions and boattail. In order for these forces to develop, something has to *react* these forces. This something is the **inertia of the rocket**, both transverse inertia and rotational inertia. The resulting forces generate a **bending moment** being applied to the rocket body. Figure 2 illustrates how the normal force acting at the nosecone (N_{NOSE}), effectively applied at the nosecone centre of pressure and the normal force acting at the fins (N_{FINS}), effectively applied at the fins centre of pressure, are reacted by the mass inertia of the rocket ($m a$) and the rotational inertia (γI_L) acting at the Centre of Gravity (CG) of the rocket. The net effect is a bending of the rocket body as shown. Note that the term $m a$ is *mass $\times$ lateral acceleration*. The Greek symbol "gamma" (γ) denotes angular (or rotational) acceleration and I_L is the rocket's longitudinal inertia that resists rotation caused by the two unequal forces, N_{NOSE} and N_{FINS} .

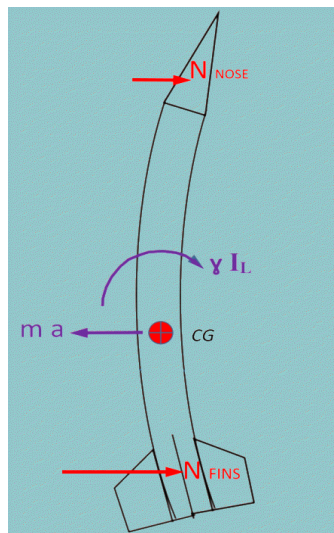


Figure 2: Bending load resulting from small angle of attack

The normal force acting on the nosecone (N_{NOSE}) is given by the dynamic pressure (q) acting on the rocket due to its flight velocity multiplied by the reference area (A), angle of attack (α) and the normal force coefficient ($C_{N\alpha N}$):

$$N_{NOSE} = q A \alpha (C_{N\alpha N}) \quad [\text{Equation 2}]$$

Similarly, the normal force acting on the fins (N_{FINS}) is given by the dynamic pressure (q) acting on the rocket due to its flight velocity multiplied by the reference area (A), angle of attack (α) and the normal force coefficient ($C_{N\alpha F}$):

$$N_{FINS} = q A \alpha (C_{N\alpha})_F \quad [\text{Equation 3}]$$

where:

q = dynamic pressure = $\frac{1}{2} \rho V^2$, lbf/in² or N/m²

A = reference area = $\frac{1}{4} \pi D^2$ where D = nosecone base diameter, in. or m.

V = vehicle velocity, ft/sec or m/s

$(C_{N\alpha})_F$ = slope of the normal force coefficient at $\alpha = 0$ ($\partial C_N / \partial \alpha |_{\alpha=0}$), per radian

α = effective angle of attack, radians

Dynamic pressure should be taken at its maximum, referred to as **Max Q**, which is the most severe combination of *velocity* and *air density*. For most EX rockets, this is simply burn out velocity and air density at burn out altitude.

For robustness of the rocket design, angle of attack used for design should be the most severe that may reasonably occur. The most severe, or *critical*, angle of attack is that which arises from either:

- Burn-out velocity combined with a horizontal wind gust of suitable magnitude. A discussion on wind velocity and how it relates to rocket structural loading is presented in [Appendix A](#)
- Dynamic imbalance (coning) owing to a combination of roll (spin) and mismatch between rocket principal axis of inertia and its geometric axis. Some degree of both roll and axis mismatch will be present for most EX rockets, despite one's reasonable effort in design and construction to eliminate such. Consequently, a pragmatic design approach is to assume that some angle of attack will arise during the critical phase of flight (Max Q) due to coning. Based on my experience, a value for alpha of 5 or 10 degrees is not out of line. In fact, owing to the natural robustness of body tubing, the penalty for such an assumption may be slight or negligible, and may preempt the dire consequence of a rocket break-up.

The normal force coefficient is calculated using the *Barrowman method*, as presented in [Appendix B](#). In order to calculate bending load, or *moment*, acting on the rocket body, it is necessary to know the inertial forces that react the applied normal loads. The inertial forces are generated solely by the mass of the rocket. At the design stage, the rocket's mass and its detail mass distribution are not known. However, a simplifying assumption can be made that the mass is evenly distributed. For most EX rockets, this is a **conservative assumption** [1], and as such, is well suited to design.

In order for this distributed mass to balance at the rocket's CG, two separate distributed loads are considered, one forward of the CG and one aft of the CG. This is illustrated in Figure 3. The location of the CG is not known, but can be taken as 1.5 or 2.0 calibres (body diameters) forward of the CP, or whatever value is chosen for the design *Stability Margin*.

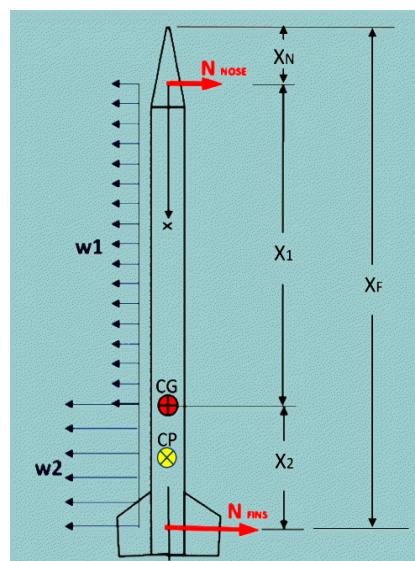


Figure 3: Aero and inertia loads acting on rocket body in flight

Distributed loads $w1$ and $w2$ are *forces* resulting from the rocket mass tending to accelerate laterally and rotationally. Equating forces and moments due to normal and inertia loads, the values of $w1$ and $w2$ can be calculated as such:

$$w_1 = \frac{N_N + N_F - w_2 x_2}{x_1}$$

[Equation 4a, 4b]

$$w_2 = \frac{N_F(2x_2 + x_1) - N_N x_1}{x_2^2 + x_1 x_2}$$

Units of w_1, w_2 are most conveniently expressed as N/mm or lb/in .

The **lateral shear (V)** as a function of x , where x is the distance along the rocket body with $x=0$ at nose CP, is given by:

$$V(x) = N_N - w_1 x \quad 0 \leq x \leq x_1$$

[Equation 5a, 5b]

$$V(x) = V_1 - w_2(x - x_1) \quad x_1 < x \leq L$$

The **bending moment (M)** as a function of x is given by:

$$M(x) = N_N x - w_1 \frac{x^2}{2} \quad 0 \leq x \leq x_1$$

[Equation 6a, 6b]

$$M(x) = V_1 x + w_2 \left(x_1 x + \frac{1}{2} L^2 - \frac{1}{2} x^2 \right) - L(V_1 + w_2 x_1) \quad x_1 < x \leq L$$

The range of x over which the two equations are valid is indicated. Units of bending moment are $N-mm$ or $lb-in$. In the equation, V_1 is the value of shear at $x=x_1$. The length L represents the body length from nose CP to the fin CP: $L = x_1 + x_2$

Figure 4 presents a cartoon of the (exaggerated) effects of both shear and bending moment on the rocket body.

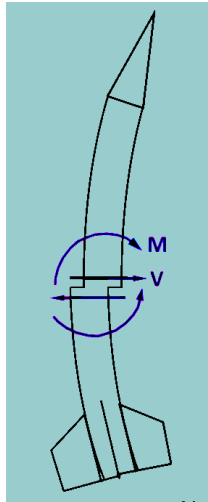


Figure 4: Effects of Shear and Bending Moment on rocket

Now that we have a suitable means of representing shear and bending moment as a function of x along the rocket body, we can generate [Shear and Bending Moment diagrams](#), which are very useful for structural design by allowing us to determine the value of shear force and bending moment at any given point along the rocket body. This provides a useful set of loads to design the rocket body, joints and any other features that affect the strength of the rocket body (for example, cutouts). An example set of Shear and Bending Moment diagrams for a typical EX rocket is given in Figure 5.

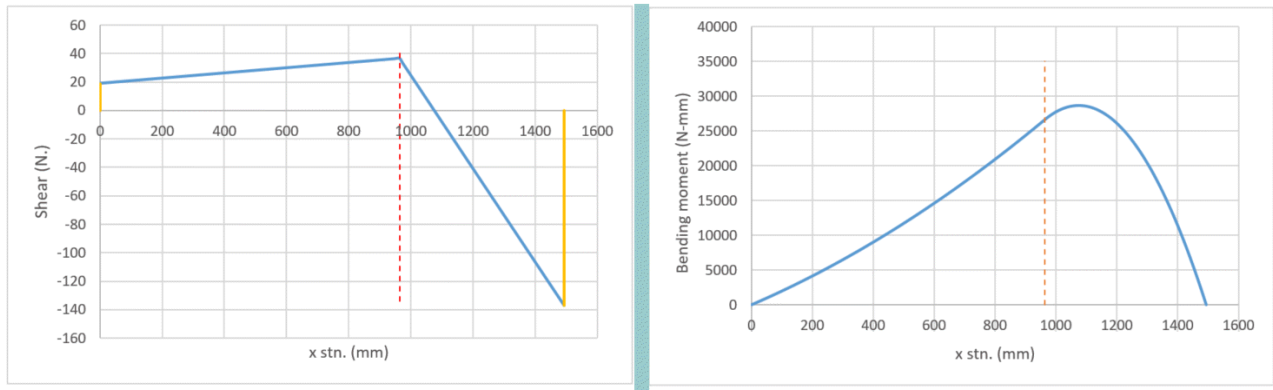


Figure 5: Example Shear and Bending Moment diagrams

For design, we are often most interested in the maximum bending moment. The maximum bending moment is given by:

$$M_{max} = \frac{N_N^2}{2 w_1} \quad 0 \leq x \leq x_1 \quad [\text{Equation 7a, 7b}]$$

$$M_{max} = \frac{V_1^2}{2 w_2} - V_1 (L - x_1) + w_2 \left(\frac{1}{2} L^2 + \frac{1}{2} x_1^2 - L x_1 \right) \quad x_1 \leq x \leq L$$

With the range of x over which the two equations are valid is indicated. The location of the maximum bending moment is where shear is equal to zero ($V = 0$). This can be seen in Figure 5.

$$x_{|V=0, M=M_{MAX}} = \frac{N_N}{w_1} \quad 0 \leq x \leq x_1 \quad [\text{Equation 8a, 8b}]$$

$$x_{|V=0, M=M_{MAX}} = \frac{V_1}{w_2} + x_1 \quad x_1 \leq x \leq L$$

Due to the inherent strength of tubular structure, shear loads can generally be neglected. Bending loads, however, can induce significant bending stress for long slender rockets. In addition to bending stress, excessive deformation of the rocket due to bending loads should be avoided, as this could lead to aerodynamic drag and dynamic stability issues.

The preceding analysis for obtaining design shear and bending loads for a rocket body can be expanded to include the effects of additional normal loading due to **body transitions** and **boattail**. This is presented in [Appendix C](#). It is also possible to assume an inertial reaction other than uniform loading, such as trapezoidal distribution, which may more accurately reflect actual mass distribution along the length of the rocket body.

The maximum **stress** due to bending of the rocket body is calculated as such:

$$f_{b \max} = \frac{M}{Z} \quad [\text{Equation 9}]$$

where **M** is the bending moment and **Z** is the **section modulus** of the body tube cross-section, based on the tube outer and inner diameters as shown in Figure 6. Units of measure can be millimetres or inches. Bending generates tensile and compressive stress, with the location of the maximum tensile and compressive stress shown.

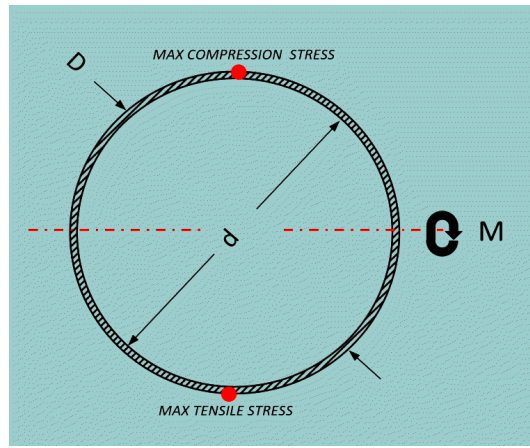


Figure 6: Cross-section of a rocket body tube

The section modulus of a tube is given by:

$$Z = \frac{\pi}{32} \frac{(D^4 - d^4)}{D} \quad [\text{Equation 10}]$$

Section modulus is a geometric property that relates to the strength of a beam. Based on the above discussion, the maximum bending stress due to worst-case aerodynamic loading can be determined for a rocket body. This maximum stress is then compared to the **strength** of the body tube material. For most EX rockets, the strength should correspond to the material's yield strength, denoted S_y , [2]. Employing the yield strength is conservative and ensures the rocket body is not permanently deformed. Alternatively, the material's ultimate strength, denoted S_u , may be used as the failure criterion. The ultimate strength corresponds to the breaking strength of the material.

If the rocket body tube is particularly lightweight and has a thin wall relative to its diameter, the tube may fail at a compressive stress level much lower than yield strength of the material. This failure mode is known as *compressive instability*, whereby the tube wall collapses, or buckles locally, at the location where compressive stress is maximum. If the rocket body diameter-to-wall thickness ratio is greater than 70 ($D/t > 70$), this failure mode needs to be considered. For example, a 3 inch diameter rocket with a wall thickness of 0.035 inch has a D/t ratio of 86. The Figure A [3] presents the critical bending strength (S_{bcr}) of a thin-walled tube based on the tube's r/t and L/r ratios. In this graph, r is the radius of the tube and L is the unsupported length of the tube. The unsupported length can be, for example, the length of the body between the nosecone and coupler. The nosecone and coupler both provide structural support to the body. This figure gives the allowable stress as the ratio S_{bcr}/E , where E is the elastic modulus of the body material. As such, this graph can be used for any body material [4]. The Figure B and Figure C presents the critical bending strength specifically for aluminum tubing.

For most EX rockets, bending of the rocket body alone may be considered sufficient for design, in particular, *LoPer* and *MiPer* class rockets. For supersonic rockets (*HiPer* and *ViPer*) where aerodynamic drag loading can be quite significant at velocities near or greater than mach 1, compressive stress on the rocket body due to aerodynamic drag should be considered. Additionally, for any class EX rocket where the acceleration is particularly high (say, $>50G$), compressive stress on the rocket body due to inertia (mass) loading should be considered. The compressive stress on the rocket body tube due to aerodynamic drag, f_{ca} , is given by:

$$f_{ca} = \frac{F_D}{A_c} \quad [\text{Equation 11}]$$

where F_D is the drag force acting on the rocket (or portion) and A_c is the body tube cross-sectional area in compression. The compressive stress on the rocket body tube due to mass inertia, f_{cm} , is given by:

$$f_{cm} = \frac{m g (1 + G_{max})}{A_c} \quad [\text{Equation 12}]$$

where m is the total mass generating the inertia, g is the gravitational acceleration and G_{max} is the maximum acceleration the rocket experiences in flight (G's). The total compressive stress is the sum of bending, drag and inertia:

$$f_{c_total} = f_b + f_{ca} + f_{cm}$$

[Equation 13]

A **Safety Factor** is then calculated based on the maximum applied stress and the *allowable* stress.

$$S.F. = \frac{S_{all}}{f_{max}}$$

[Equation 14]

Where the allowable stress, S_{all} , is either S_y , S_u or S_{bcr} and the maximum applied stress, f_{max} , is either f_b , f_c or f_{c_total} , whichever is appropriate. Value of the Safety Factor should be at least two for metallic bodies and three or greater for rocket bodies made of non-metallic material such as PVC or glass-reinforced plastic (GRP).

Body Joints

In a perfect world, a rocket body would consist of a single tube extending the full length from the nosecone to the fins. In reality, a rocket body will consist of two or more sections joined together (at a *joint*). This is usually done to facilitate the recovery system, which involves "breaking" the rocket into two sections to release a drogue and/or main parachute. This can also be the case whereby the rocket motor serves as airframe, and is joined to the forward, or payload, section.

If a joint is not properly designed, it could serve as a weak link. A joint must possess sufficient *strength* to take bending loads (in flight and handling/transport) and must be *rigid* such that it does not allow the rocket to flex at the joint. Flexing, combined with dynamic effects in flight, could lead to breakup of the rocket.

The simplest joint design is a length of tubular *Coupler* that slides neatly within the two sections of body tubes to be joined, as illustrated in Figure 7. This design offers both good strength and rigidity. If the joint tube is made of the same material and same wall thickness, and has a length of twice the body diameter, strength of the joint can be reasonably close to the strength of the original body tube. With careful design (combination of joint tube wall thickness and length) strength of the body tube can be matched (although this is generally overkill).

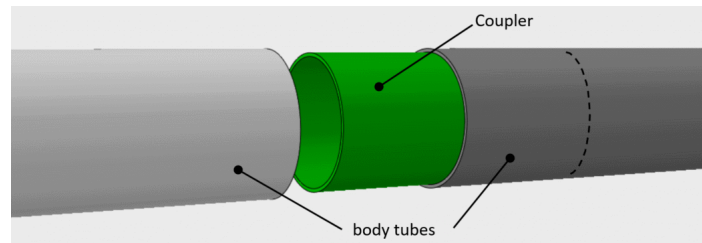


Figure 7: Example of a joint using a coupler tube

As mentioned earlier, bending stress is proportional to the section modulus (Z) of a body tube cross-section, given by:

$$Z = \frac{\pi}{32 D} (D^4 - d^4)$$

[Equation 15]

Assuming a coupler tube has the same outer diameter as the inner diameter of the body tube, and the body tube is considered to be a thin-walled structure, the **coupler wall thickness** (t) is given by:

$$t = \frac{1}{2} d - \left(\frac{d^4}{8} - \frac{D^4}{16} \right)^{1/4}$$

[Equation 16]

Units of measure can be millimetres or inches. With this wall thickness of the coupler, the bending strength matches that of the rocket body tube (assuming they are made of the same material and the tube does not buckle when subjected to bending). For example, a rocket with a body tube outer diameter of $D=75\text{mm}$ and inner diameter $d=71.7\text{mm}$ (i.e. wall thickness is 1.65mm), the coupler thickness calculates to a value of $t = \underline{1.92\text{mm}}$.

This type of joint serves well as a *separable joint*, such as that required to separate a rocket for release of a parachute. In such a case, nylon screws are used to hold the joint together. Nylon screws have a low shear strength that allows the joint to separate when the separation charge is fired. This will be discussed in detail in the [Recovery System](#) Design webpage.

The possible drawbacks to this type of joint is that it occupies length and can be rather heavy, especially for larger EX rockets. For a *non-separable joint*, a better option is a *Ring Coupler*, as shown in Figure 8.

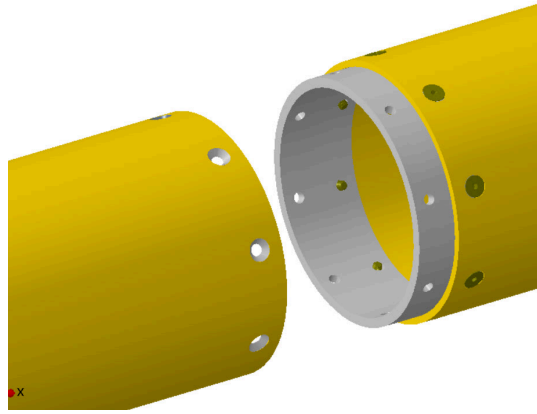


Figure 8: Example of a joint using a ring coupler

With a ring coupler joint, any bending loads (or *bending moment*) that the rocket is subjected to is carried by the coupler attachment screws in shear. The advantage of a ring coupler is the short length it occupies and can be very lightweight. The holes in the coupler are tapped to accommodate the joint screws. The minimum wall thickness of the coupler is determined in the same manner as the Tube Coupler. It may be necessary to make the wall with additional thickness to provide enough thread length such that the threads are not inadvertently stripped due to repeated installation and removal of the joint screws. The minimum number of screws in the joint should be six, with typically eight or more being used for larger diameter rockets.

The maximum shear force (F_{max}) that any of the attachments screws is subjected to is dependent upon the total number of fasteners in the ring coupler joint:

$$F_{max} = f \frac{M}{D} \quad [\text{Equation 17}]$$

where:

F_{max} = maximum force (lbf or N.) tending to shear the fastener

f = load fraction dependent upon number of fasteners in joint

- $f = 2/3$ for six fasteners
- $f = 1/2$ for eight fasteners
- $f = 2/5$ for ten fasteners
- $f = 1/3$ for twelve fasteners

M = applied bending moment (lbf-inch or N-m)

D = outer diameter of coupler (inches or mm)

The edge distance (e) of the fasteners should be no less than $2x$ the diameter of the hole (D) in the part to prevent *shear tear-out* failure ($e = 2D$, minimum). For countersunk fasteners, "D" should be taken as the *average* hole diameter.

[Click for illustration.](#)

With regard to fasteners, it is recommended to use either [Alloy Steel Flat Head Cap Screws](#) or [Alloy Steel Button Head Cap Screws](#) manufactured per ASTM F823 or ASTM F823M. These are available in both SAE and metric sizes. One important advantage to using these fasteners, compared to regular COTS [\[5\]](#) fasteners, is that these have a guaranteed strength. A second advantage is that these alloy steel screws have excellent strength and, as such, surprisingly tiny fasteners can be used for applications such as coupler attachment. The alloy steel has a tensile strength of 145 ksi (1000 MPa) and shear strength of 95 ksi (655 MPa). This compares to typical COTS fasteners that may have a tensile strength of 50-60 ksi (345-415 MPa) and shear strength of 35-40 ksi (240-275 MPa).

[Table of Screw Strength, SAE](#)
[Table of Screw Strength, Metric](#)

Ring Coupler joints may also fail if the fastener *bearing stress* is too high. Bearing stress is defined as the fastener shear load acting over the contact area. As such, bearing stress is the *contact pressure* that results

when a joint is subjected to a bending moment.

$$f_{br} = \frac{F_s}{D_m t}$$

[Equation 18]

where:

F_s = shear force acting on the fastener (lbf or N.)

D_m = fastener major diameter (inches or mm)

t = wall thickness of the body tube at the joint (inches or mm)

If the bearing stress is too high, the joint will fail by [shear-tear out or by gross elongation](#) (ovalization) of the hole in the body tube. The bearing stress must be lower than the bearing strength of the body tube material, with the desired safety factor applied.

$$f_{br} \leq \frac{S_{br}}{S.F.}$$

[Equation 19]

where:

S_{br} = Ultimate bearing strength of body tube material (psi or N/mm².) [Example, for 6061 tube](#)

Resources:

Res.4 [Metallic Materials Properties Development and Standardization](#) issued by the Federal Aviation Administration (FAA) recognized internationally as a reliable source of metals strength data for aerospace design

Res.5 [Rocket vehicle loads and airframe design](#) Aspirepace technical papers Authors: Rick Newlands, Martin Heywood, Andy Lee

Res.6 [Metallic Materials and Elements for Aerospace Vehicle Structures](#) MIL-HDBK-5H, a public domain reference for aerospace design data for metallics.

Notes:

[1] *Conservatism* is a design or analysis approach commonly used in aerospace engineering to help simplify a problem by using simpler or less detailed analysis methods or techniques that errs on the side of a higher factor of safety.

[2] Material strength (stress) allowables, such as yield or ultimate strength, are usually denoted with the letter F , such as F_y and F_u . To avoid potential confusion, the symbol F will be used solely to denote *force* (such as F_d for drag force) rather than *stress*. The symbol S is used here to denote an allowable stress. However, the convention of using lower case f to represent applied stress will be maintained.

[3] This graph is based on Figure C8.13 of *Analysis and Design of Flight Vehicle Structures*, E.F.Bruhn

[4] This graph was developed based on strength testing of aluminum and steel tubes. As such, due diligence should be taken when using this graph for materials other than these. For example, ensuring a healthy safety factor is obtained, or by proof-load testing the rocket under bending load.

[5] COTS = Commercial Off The Shelf

[Derivation](#) of equations for w_1 and w_2 assuming uniformly distributed loading due to inertia

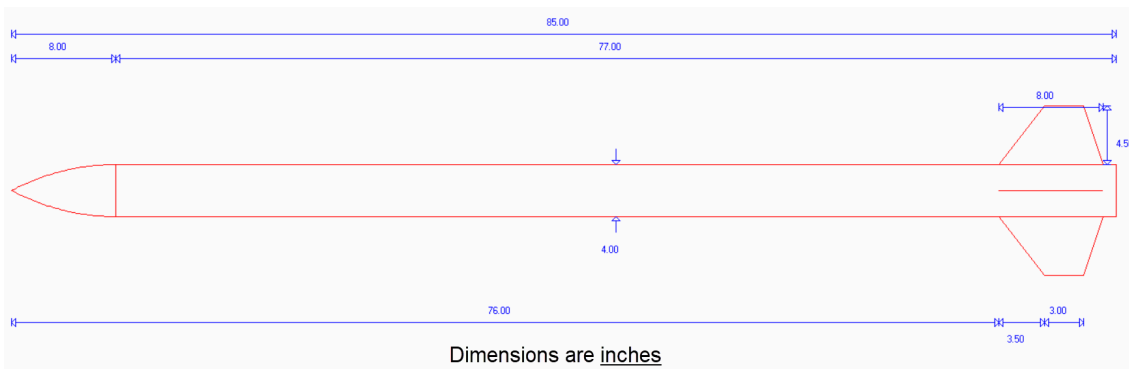
[Derivation](#) of equations for Shear and Bending Moment assuming uniformly distributed loading due to inertia

[Derivation](#) of equation for fastener shear force for 6 fasteners.

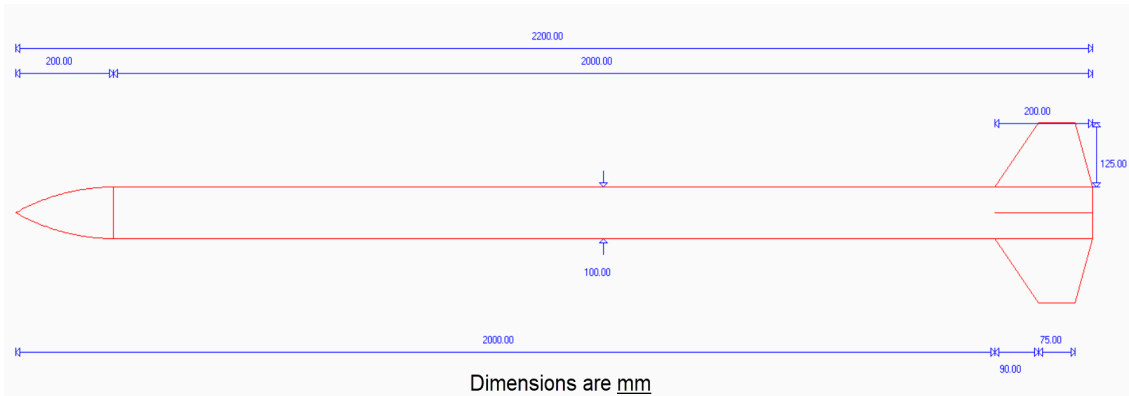
[Derivation](#) of equation for fastener shear force for 10 fasteners.

Examples

[Example 1](#) - Check body structural strength for the following proposed EX rocket design (U.S./imperial units):



Example 2 - Check body structural strength for the following proposed EX rocket design (metric units):



[Next--Nosecone Design](#)

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[Return to Rocket Design Index page](#)

[Return to Home Page](#)
